Rejection Sampling for Medical MCQ Generation and Fine-Tuning TinyLlama for Layperson QA

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PowerPoint Slides: Google Drive PDF
Video Presentation: Panopto Video
Code Repository: github.com/chelseanbr/aih-project
Data Source: huggingface.co/datasets/lavita/MedQuAD

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Abstract

Access to accurate, understandable medical information is essential, especially for laypersons without clinical expertise. This project explores how to improve layperson medical question answering (QA) by generating multiple-choice questions (MCQs) from the MedQuAD dataset and fine-tuning TinyLlama using a lightweight, parameter-efficient method (LoRA). Through rejection sampling and few-shot Chain-of-Thought prompting, a small MCQ dataset was curated. Fine-tuning led to some gains over baseline TinyLlama and MedAlpaca models, demonstrating the potential of focused domain adaptation even on smaller models.

1 Introduction

Medical QA remains a tough challenge for large language models (LLMs). Medical language is complex, and layperson users require responses that are both medically accurate and easy to understand. Many general-purpose LLMs are not reliably tuned for this domain, leaving significant gaps in accessibility to trusted medical information. To address this, I developed a pipeline to generate high-quality MCQs from the MedQuAD dataset and fine-tuned TinyLlama to better handle layperson medical QA. Improving accessibility to medical QA supports broader public health goals by making accurate, understandable answers more available to everyone.

2 Related Work

Ben Abacha and Demner-Fushman [1] developed an entailment-based framework for medical QA, focusing on ensuring that answers are both relevant and well-justified. Their work led to the creation of the MedQuAD dataset, which forms the foundation for the layperson question-answer pairs used in this project. Yuan et al. [2] proposed a method where a pre-trained LLM is guided through in-context learning to generate a dataset of high-quality question, reasoning, and answer triples. This idea of using model-generated reasoning to enhance training data quality influenced my approach, where I incorporated Chain-of-Thought prompting during MCQ generation to encourage more structured, explainable answers. Zhu et al. [3] demonstrated how synthetic QA data generation, when carefully controlled, can boost medical QA model performance. This project builds on these ideas by combining curated QA generation with efficient fine-tuning for a small, adaptable model.

3 Methodology

The project follows a two-stage pipeline: MCQ generation and TinyLlama fine-tuning. The overall process is shown in Figure 1.

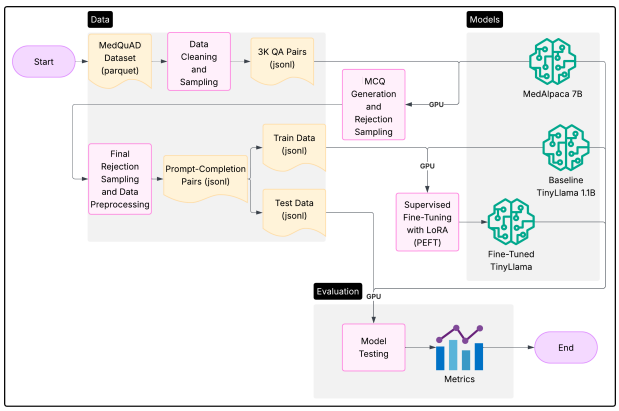

Figure 1: Workflow for MCQ Generation and Fine-Tuning

```
# Few-shot example to encourage structure
few_shot_example = """
Question: What is asthma?
Answer: Asthma is a lung condition that causes airway inflammation and breathing difficulty.
<choices>
A. A skin condition
B. A viral infection
C. A lung disease that causes airway inflammation
D. A type of heart failure
</choices>
<reason>Asthma is a lung condition that causes airway inflammation and breathing difficulty.</reason>
<answer>C</answer>
"""

# Format QA into prompt
def format_prompt(question, answer):
    return f"""
Create layperson multiple-choice answer choices and reasoning for the given medical Q&A pair.
Include 4 total choices: one correct and three plausible, but incorrect. Follow the XML-style structure below:
{few_shot_example}
Now it's your turn:
Question: {question}
Answer: {answer}
"""
```

Figure 2: MCQ Generation Prompt

3.1 MCQ Generation

I began by sampling 3,000 QA pairs from the MedQuAD dataset. A custom few-shot prompt (Figure 2) with a Chain-of-Thought (CoT) example was designed to guide MCQ generation. Few-shot prompting helped the model produce well-structured questions with reasoning steps.

MedAlpaca was used to generate the MCQs on Google Colab, which took about 1.5 hours on an A100 GPU. Three responses were generated for each question-answer pair, then filtered using rejection sampling to ensure the outputs were medically accurate, clear, and well-formatted. The prompt was designed to elicit a specific format for the resulting MCQs, including a question, answer choices, an explanation of the correct answer, and the correct answer inside tags. Previews of the sampled and generated data are shown in Figure 3 and Figure 4.

```
{"question":"What causes Gum (Periodontal) Disease?","answer":"Gum disease is caused by dental plaque -- a sticky film of bacteria that builds up on teeth. Regular brushing and flossing help get rid of plaque. But plaque that is not removed can harden and form tartar that brushing doesn't clean. Only a professional cleaning by a dentist or dental hygienist can remove tartar."}
```


Figure 3: Example of Sampled Data from MedQuAD


```
{"question":"What causes Gum (Periodontal) Disease?","mcq_choices":"A. Genetic mutations\nB. Poor dental hygiene\nC. Injury to the gums\nD. Periodontal abscesses","mcq_reason":"Gum disease is caused by dental plaque -- a sticky film of bacteria that builds up on teeth. Regular brushing and flossing help get rid of plaque. But plaque that is not removed can harden and form tartar that brushing doesn't clean. Only a dentist or dental hygienist can remove tartar.","mcq_answer":"B"}
```


Figure 4: Example of Generated MCQ after Prompting


```
{"prompt":"You are a helpful and concise medical assistant.\n\nAnswer the following multiple-choice question. First, briefly explain your reasoning using relevant medical knowledge. Then state your final answer letter using exactly one <answer></answer> tag.\n\nInstructions:\n- Use only one <answer> tag (e.g., <answer>B</answer>)\n- Choose the best answer letter\n- Stop immediately after the </answer> tag\n\n<question>\nWhat causes Gum (Periodontal) Disease?\nA. Genetic mutations\nB. Poor dental hygiene\nC. Injury to the gums\nD. Periodontal abscesses\n</question>\n","completion":"Gum disease is caused by dental plaque -- a sticky film of bacteria that builds up on teeth. Regular brushing and flossing help get rid of plaque. But plaque that is not removed can harden and form tartar that brushing doesn't clean. Only a dentist or dental hygienist can remove tartar.\n<answer>B</answer>"}
```


Figure 5: Example of Finalized Data for Train/Test

Following manual filtering, 511 MCQs were finalized (Figure 5), with an 80%/20% train/test split.

3.2 Fine-Tuning TinyLlama with LoRA

TinyLlama was fine-tuned using LoRA (Low-Rank Adaptation), which inserts small trainable matrices into attention layers instead of updating the full model. This enables efficient adaptation with limited resources.

Fine-tuning parameters used included:

- LoRA Rank: 8
- LoRA Alpha: 32
- Optimizer: AdamW, learning rate 1e-3
- Batch Size: 16
- Epochs: 3
- Weight Decay: 0.01
- Mixed Precision: FP16
- Gradient Checkpointing: Enabled

The model was trained using supervised learning to predict the correct MCQ answer label.

4 Results

4.1 Answer Distributions

Figure 6 confirms that the train/test splits were balanced. Figure 7 shows that baseline TinyLlama was biased toward a single answer choice. Fine-tuning improved answer distribution and reduced missing answers (Figure 8), while MedAlpaca (Figure 9) did better than the baseline, but often skipped questions entirely.

4.2 Loss and Accuracy

Since the dataset was small, training completed within a few epochs. The training and validation loss (Figure 10) was a bit noisy and the model was prone to overfitting, but overall showed improvement, even with limited data.

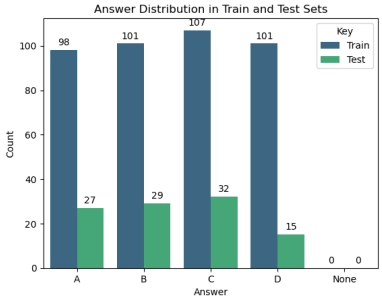

Figure 6: Train/Test Answer Distribution

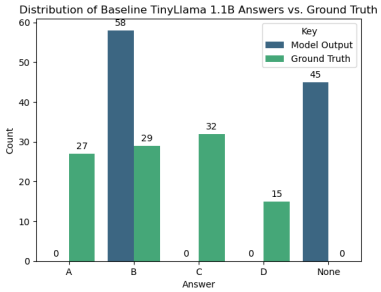

Figure 7: Baseline TinyLlama Answer Distribution

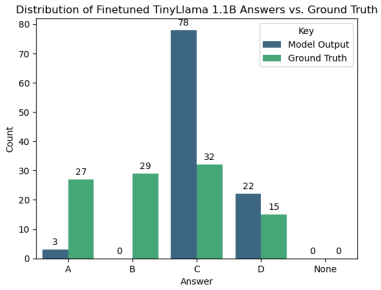

Figure 8: Fine-Tuned TinyLlama Answer Distribution

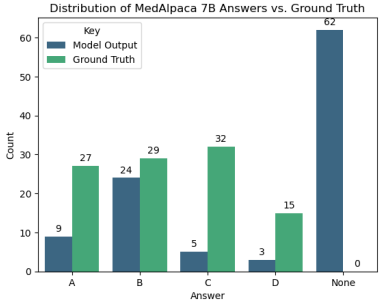

Figure 9: MedAlpaca Answer Distribution

[48/48 06:11, Epoch 3/3]

Epoch	Training Loss	Validation Loss
1	0.846100	0.489880
2	0.370500	0.475589
3	0.363300	0.481885

Figure 10: Training and Validation Loss

As shown in Figure 11, fine-tuned TinyLlama achieved 33.98% accuracy, improving over the 14.56% baseline and 18.45% MedAlpaca.

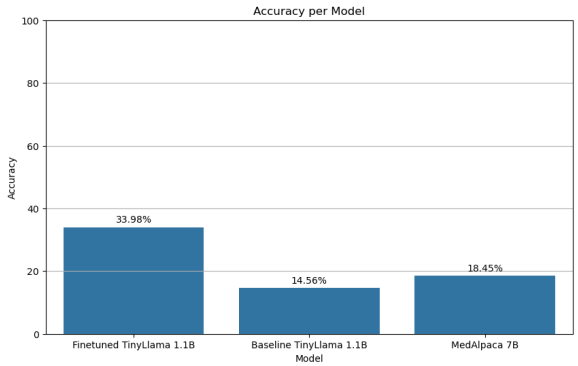

Figure 11: Accuracy Comparison Across Models

5 Conclusion

This project built a full pipeline for improving layperson medical question answering by combining rejection-sampled MCQ generation with lightweight LoRA fine-tuning on TinyLlaMA. Even with a small, curated dataset and a compact model, fine-tuning boosted accuracy over both the baseline TinyLLaMA and MedAlpaca. While the gains were not huge, they showed that even small models can get noticeably better with focused domain adaptation.

If I had more time, I would scale this up by fine-tuning larger models like LLaMA-2 and running stronger comparisons against GPT-based models. I would also try smarter ways to generate MCQs, like using reinforcement learning or contrastive prompting, instead of just filtering outputs with rejection sampling. Finally, I would move beyond multiple-choice to open-ended QA tasks, and test across more diverse medical datasets to better understand how well the model can generalize to real-world questions.

References

[1] Asma Ben Abacha and Dina Demner-Fushman. 2019. A Question-Entailment Approach to Question Answering. *BMC Bioinformatics* 20, 1 (2019). https://bmcbioinformatics.biomedcentral.com/articles/10.1186/s12859-019-3119-4 Article number 511:1–511:23.

[2] Zheng Yuan, Hongyi Yuan, Chengpeng Li, Guanting Dong, Keming Lu, Chuanqi Tan, Chang Zhou, and Jingren Zhou. 2023. Scaling Relationship on Learning Mathematical Reasoning with Large Language Models. *arXiv preprint arXiv:2308.01825* (2023). https://arxiv.org/abs/2308.01825

[3] Jiean Zhu. 2024. Cura-LLaMA: Evaluating Open-Source Large Language Model's Question Answering Capability on Medical Domain. *Applied and Computational Engineering* 90 (2024), 52–60. doi:10.54254/2755-2721/90/20241725